Chapter One

Introduction to Data Communications

and

computer networks

Compiled by Yilkal B.

Chapter Outline

- ✓ *Basics of data communications*
- ✓ *Basic components of communication network*
- ✓ *Current and future networks*
- ✓ *Fundamentals of Information Transmission Media and Coding*
- ✓ *Types of networks [PP &CS]*
- ✓ *Basic Data Communication Hardware [NIC, cables, Switch, Router, Hub, Bridge…]*
- ✓ *Basic types of network Topologies: ring, bus, star, and mesh*

Compiled by Vikash P

Introduction

- ***Information:*** *defined as a* ***collection of facts*** *from which conclusions may be drawn is an important resource*
- *The need of information has increased from time to time*
  - *leads to the need of sharing of information among different agents*
- ***Data communication*** *is the exchange of information between two agents*
- ***Old paradigm:***
  - *A single powerful computer serving all the needs of an organization*
  - ***Sneaker-net*** *is Method of sharing data by copying it to a disk and carrying it from computer to computer*

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- *New paradigm*
  - ***Computer networks:*** *a large number of separate (autonomous) but internetworked (being able to exchange information) computers doing the job*
    - *Merging of computer and communications technologies – no geographical barrier*
    - ***Connection:*** *copper wire, fiber optics, microwaves, infrared, communication satellites, …*
- *Definition:*
  - ***A computer network*** *is an interconnected collection of autonomous computers.*

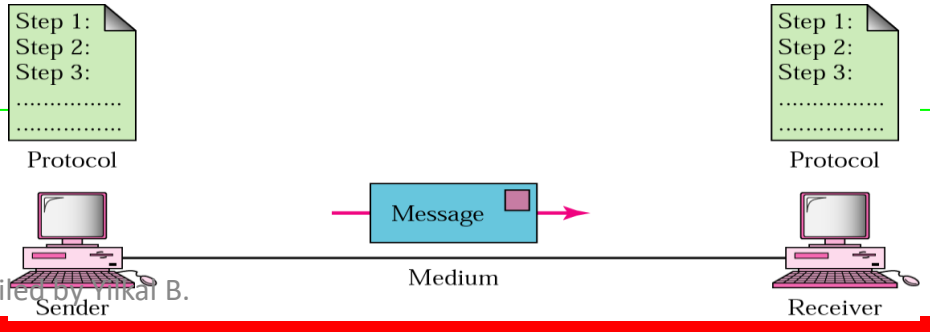
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- ***Interconnected*** *meaning two computers have the ability to exchange information using some transmission media.*
- ***Autonomous*** *meaning where no one computer controls any other computer (i.e. no computer can forcibly start or stop another computer)*
- ***Computers*** *can be PC's, workstations and other "specialized" computers such as hubs, switches and routers*
- *The computers can be geographically located anywhere*

Overview of Data Communications

- *A data communication system has **5 components***

1. ***Message:*** *the information to be communicated (text, numbers, pictures, sound, video - or combinations)*
2. ***Sender:*** *the device - computer, video camera, …*
3. ***Receiver:*** *still the device*
4. ***Medium:*** *the physical path by which a message travels from sender to receiver*
5. ***Protocol:*** *the set of rules that govern data communications; **an agreement** between the communicating devices.*

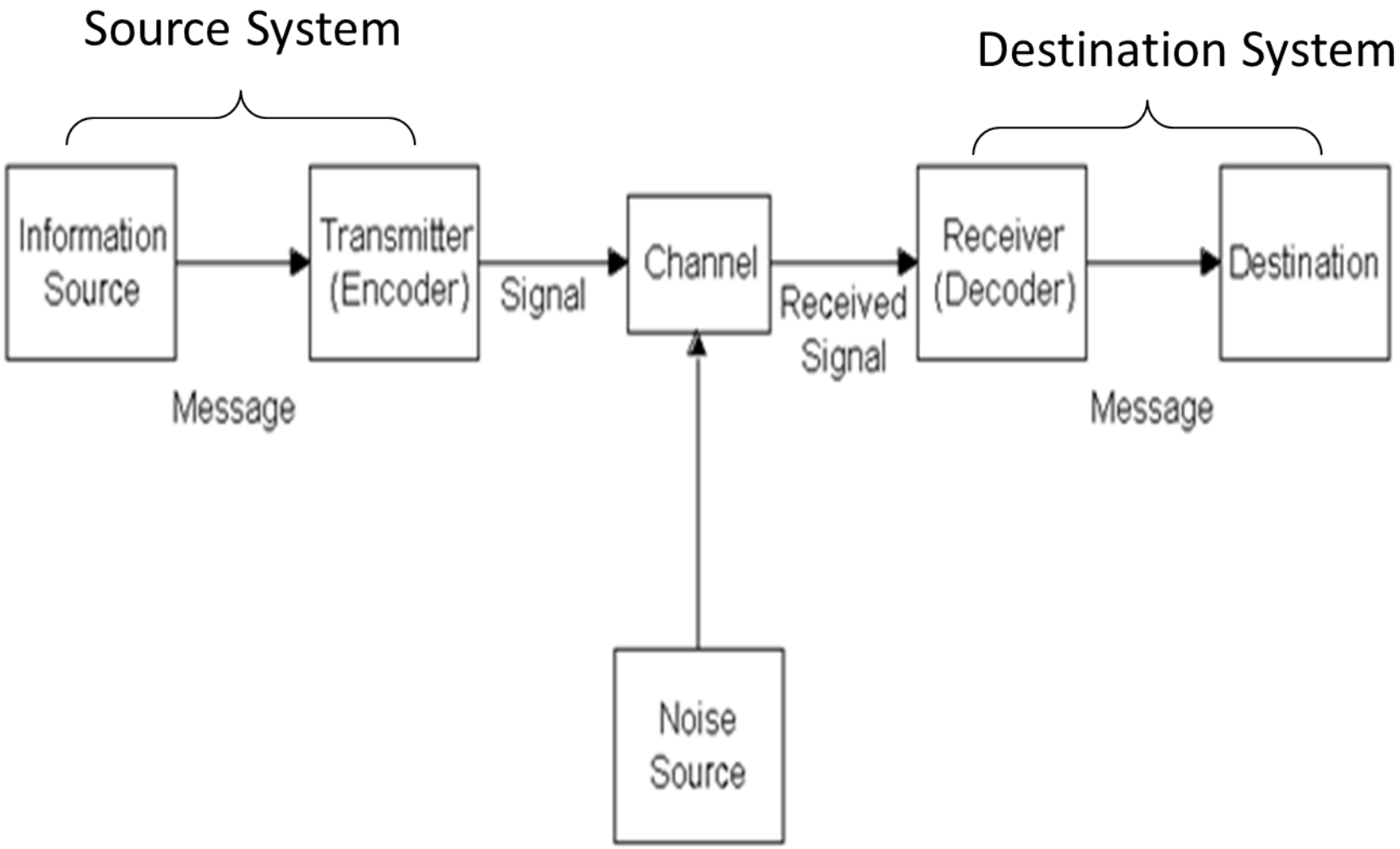


Communication Model

❖ ***The key elements of the model are:***

- ***Source:*** *This device* ***generates*** *the data to be transmitted; examples are telephones and personal computers.*
- ***Transmitter:*** ***transforms and encodes the information*** *in such a way as to produce electromagnetic signals.*
- ***Transmission System:*** *This can be a* ***single transmission*** *line or a complex network connecting source and destination.*
- ***Receiver:*** *The* ***receiver accepts the signal from the transmission system*** *and converts it into a form that can be handled by the destination device.*
- ***Destination:*** *Takes the incoming data from the receiver*

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Communication Model

Some Key Of Communications Tasks

- ***Transmission System Utilization:*** *the need to make efficient use of transmission facilities that are typically shared among a number of communicating devices*
  - *Multiplexing*
  - *Congestion control techniques*
- ***Signal Generation:*** *The properties of the signal*
  - *capable of being propagated through the transmission system*
  - *Interpretable as data at the receiver*
- ***Error detection and correction:*** *In all communications systems, there is a potential for error that should be detected and corrected.*

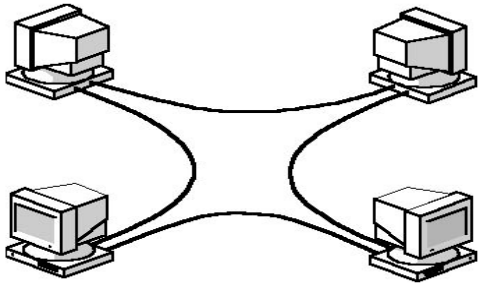
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- **Addressing and routing:** a source system must somehow indicate the identity of the intended destination
  - A specific route must be chosen from many alternative routes
- **Flow control:** required to assure that the source does not overwhelm the destination by sending data faster than they can be processed and absorbed
- **Protocols:** Rules that govern data communication, including error detection, message length, and transmission speed
- **Attenuation:** Loss of power in a signal as it travels from the sending device to the receiving device

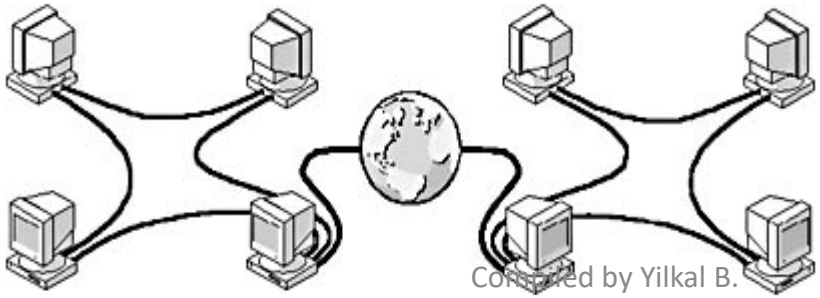
Network Categories

- *Based on **size**, **ownership**, **the distance it covers***

- ***Local Area Network (LAN):*** *usually privately owned and links devices in a single office, building or campus.*



- ***Wide Area Network (WAN):*** *covering large geographic area; may utilize public, leased, or private communications equipment*



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- ***Metropolitan Area Network (MAN):*** *designed to extend over an entire city; it may be a single network or interconnected LANs*
- ***Personal Area Network (PAN):*** *meant for one person. e.g. a wireless network connecting a computer with its mouse, keyboard and printer.*

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

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❖ **Network Architecture:** includes the type of computers on the network and determines how network resources are handled.

❖ **Two common types** (**Peer to Peer** and **Client to Server**)

A. Peer-to-peer: Each node considered as equal in terms of resource sharing and responsibilities

❖ **pros**

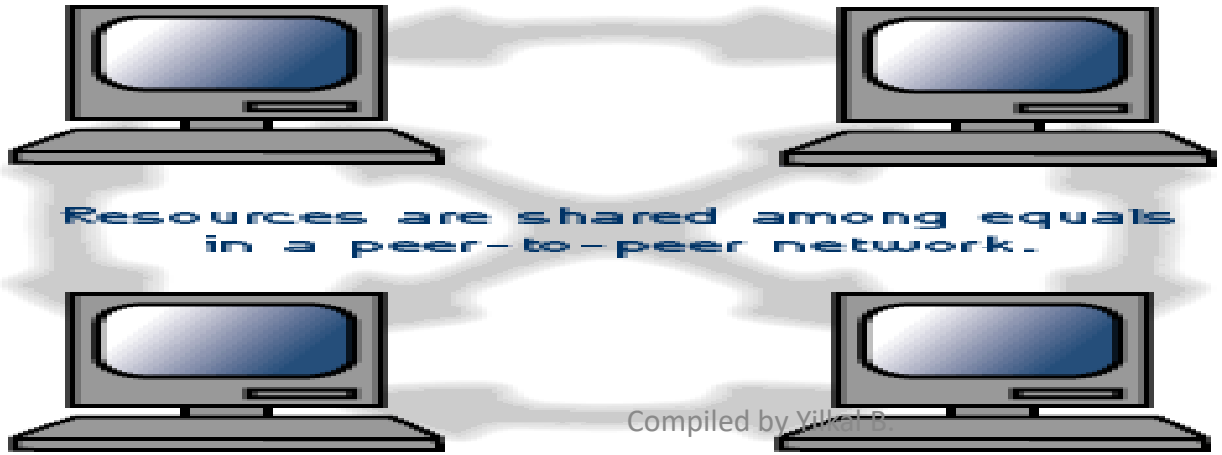
- Easy to set up
- Less expensive
- Demands moderate level of skill to administer
- User is able to control their own resources

❖ **Cons**

- Only < 10 nodes
- Very low level o security
- Performance suffers when a computer is accessed

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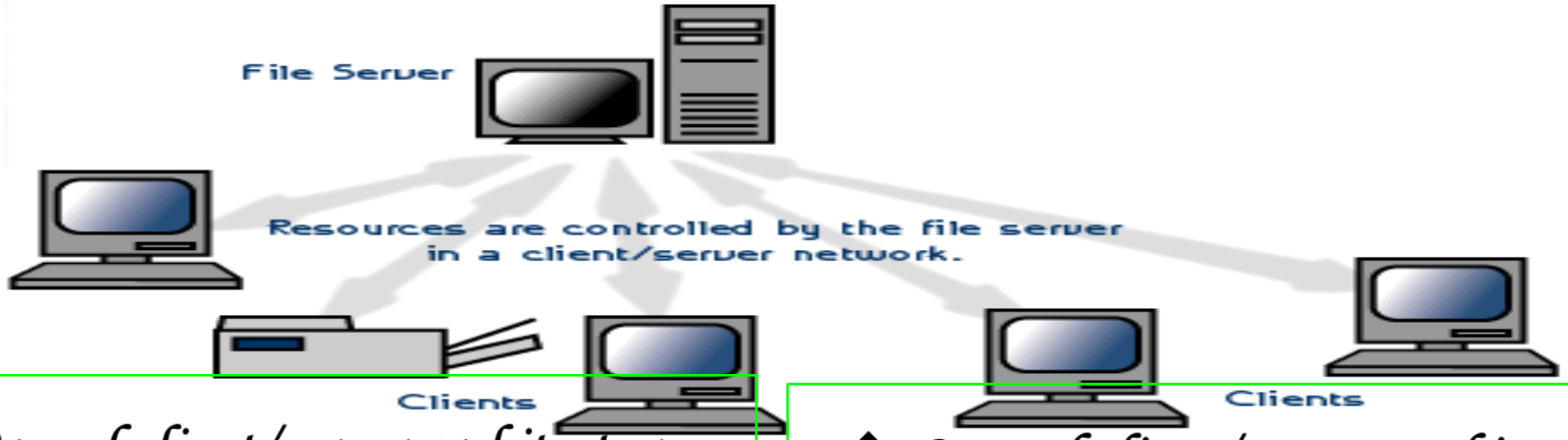
- *Peer-to-peer networks are **good choices for environments where**:*
  - *There are 10 users or fewer*
  - *Users share resources, such as printers, but no specialized servers exist*
  - *Security is not an issue*
  - *The organization and the network will experience only limited growth within the foreseeable future*



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B. Client to Server Model: *the client request for a service and the server deliver the resource(resourceful).*

- *Consists of a group clients connected to a server*
- *Server – with more RAM, larger hard disk, more processing power…*



❖ *Pros of client/server architecture*

- *Sharing Resources*
- *Security*
- *Can support thousands of users*

❖ *Cons of client/server architecture*

- *more complex to install, configure, and manage*
- *Expensive*

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❑ Servers in networking

1. **File and Print Servers:** File and print servers manage user access and use of file and printer resources.

2. **Application Servers:** Application servers make the server side of client/server applications, as well as the data, available to clients.

3. **an application server** differs from a file and print server. With a file and print server, the data or file is downloaded to the computer making the request. With an application server, the database stays on the server and only the results of a request are downloaded to the computer making the request.

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3. Mail Servers: *Mail servers operate like application servers in that there are separate server and client applications, with data selectively downloaded from the server to the client.*

4. Fax Servers: *Fax servers manage fax traffic into and out of the network by sharing one or more fax modem boards.*

5. Directory Services Servers: *Directory services servers enable users to locate, store, and secure information on the network.*

Network Topology

- **Topology** refers to the way in which multiple devices are interconnected via communication links.
- There are two types of topology: physical and logical.

Physical Topology

- Refers to the way in which a network is laid out physically
- Refers to the arrangement or physical layout of computers, cables, and other components on the network
- Can be referred as Physical layout, Design, Diagram, Map
- Two or more devices connect to a link; two or more links form a topology

Logical topology: is bound to network protocols and describe how data is moved across the network

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- *A network's topology affects its capabilities*
- *The choice of one topology over another will have an impact on:*
  - *The types of equipment that the network needs*
  - *The growth of the network – scalability*
  - *The way the network managed*
- *Four basic topologies are possible:* ***bus, star, ring, mesh, and hybrid***

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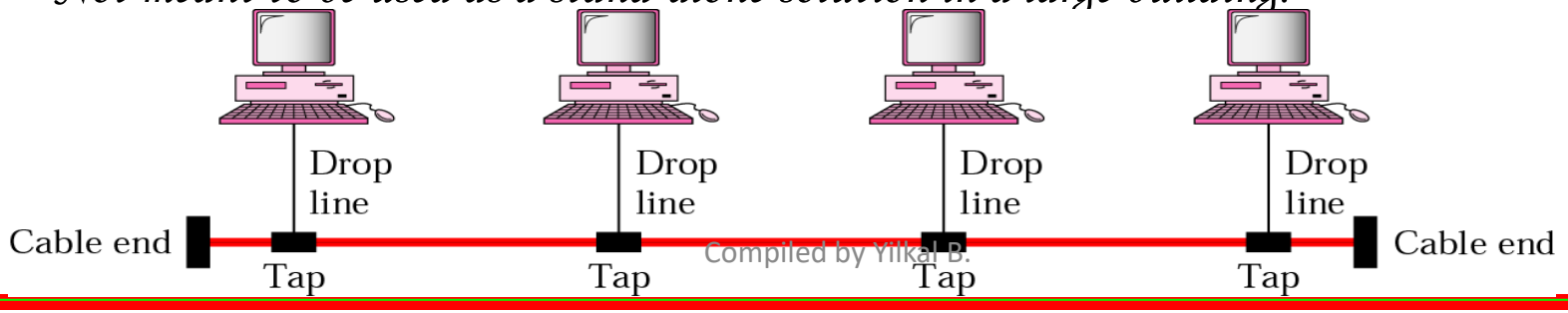
1. Bus: *multipoint (one long cable acts as a backbone to link all the devices in the network)*

Advantages

- *Easy to connect a computer or peripheral to a bus.*
- *Requires less cable length than a star topology.*

Disadvantages

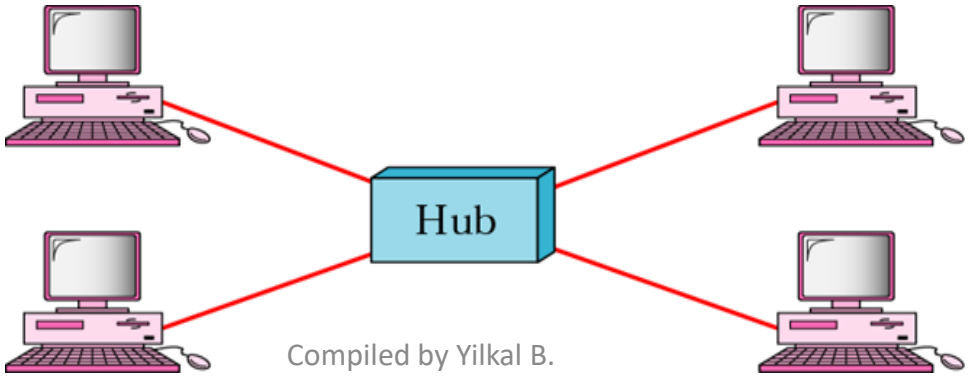
- *Entire network shuts down if there is a break in the main cable.*
- ***Terminators** are required at both ends of the backbone cable.*
- *Difficult to identify the problem if the entire network shuts down.*
- *Not meant to be used as a stand-alone solution in a large building.*



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2. Star: *each device has a dedicated point-to-point link only to a central controller, usually called a Concentrator*

- ***Advantages***
  - *robust; a failure of a link has no effect on others*
  - *fault identification and isolation are easy*
  - *less expensive than mesh (but more expensive than others)*
- ***Disadvantage***
  - *Single point of failure*
  - *Requires more cable length than a linear topology*
  - *More expensive than linear bus topologies because of the cost of the concentrators.*



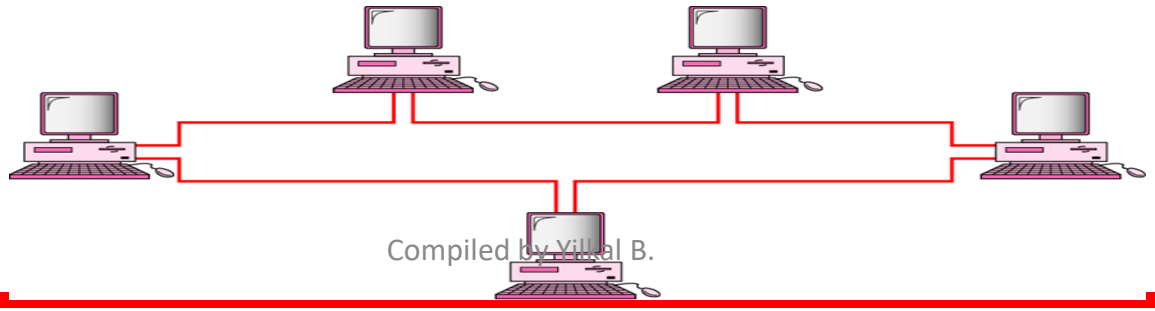
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3. Ring: *each device has a dedicated point-to-point connection only with the two devices on either side of it.*

- *a signal is passed along the ring in one direction, from device to device, until it reaches its destination*
- *each device incorporates a repeater*

- ***Advantages***
  - *Equal access for all users*
  - *Each workstation has full access speed to the ring*
  - *As workstation numbers increase performance diminishes slightly*

- ***Disadvantages***
  - *Costly Wiring*
  - *Difficult Connections*
  - *Expensive Adaptor Cards*



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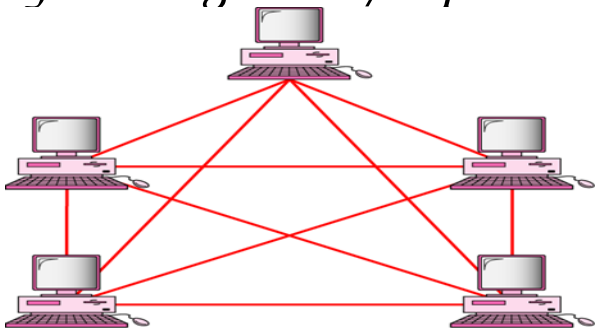
4. Mesh: *every device has a dedicated point-to-point link to every other device*

- *every device must have n-1 I/O ports*

➢ ***Advantages***

- *no traffic problem*
- *robust; a failure of a link has no effect on others*
- *privacy or security*
- *fault identification and isolation are easy*

➢ ***Disadvantages:*** *amount of cabling and I/O ports needed (expensive)*



5. Hybrid topology: *A combination of two network topologies.*

- *E.g. star and bus*

Internetworking

- *Interconnection among or between public, private, commercial, industrial, or governmental networks Called also internet*
- *Three variants*
  - *Intranet*
  - *Extranet*
  - *Internet*
  - *internet*
- ***Intranet:*** *a set of networks that is under the control of a single administrative entity*
- ***Internet:*** *Global network of networks*
  - *worldwide interconnection of networks*
- ***internet*** *–refers to bridged network in general*
- ***Extranet:*** *is a private network similar to an intranet, but typically open to external parties, such as business partners, suppliers, key customers, etc*

Mode of transmission

- refers to the direction of signal flow between two linked devices
- It can be
  - **Simplex:** unidirectional, only one of the devices can transmit

    E.g. TV transmission, radio
  - **Half-duplex:** both can transmit and receive, but not at the same time

    E.g. wireless handset (walkie -Talkie)
  - **Full-duplex:** both can transmit and receive at the same time

    E.g. Telephone transmission

Transmission media

- *Is a **physical media** that carries a signal from the transmitter to the receiver*
- *The information or signal transmitted from one device to another is through electromagnetic signals.*
- ***Electromagnetic signals** include power, voice, radio, waves, infrared light, visible light, ultraviolet light, X-rays, and gamma rays.*
- *These signals can travel trough **vacuum**, **air** or any other transmission medium*
- *The measurement of the quantity of data that can be passed down(transmitted) a communication link in a given time is done in terms of bandwidth (**bits per second**).*

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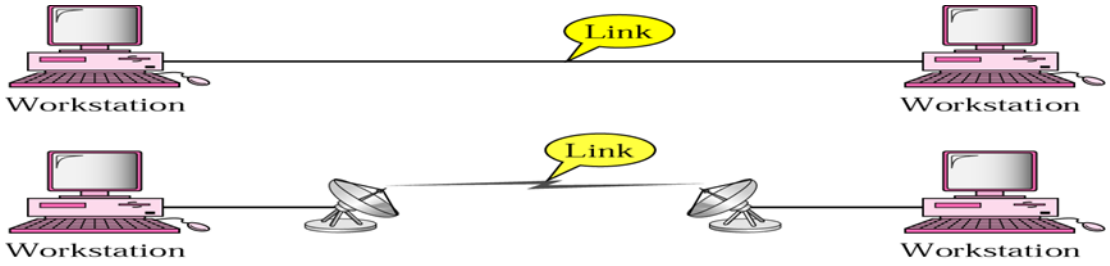
- *Two basic categories*
  - *Guided*
  - *Unguided*
- ***Guided:*** *uses a cabling system that guides the signals along a specific path.*

  E.g. Fiber Optics, Twisted Pair, Coaxial Cable etc.…
- ***Unguided:*** *consists of a means for the data signals to travel but nothing to guide them along a specific path - wireless .*

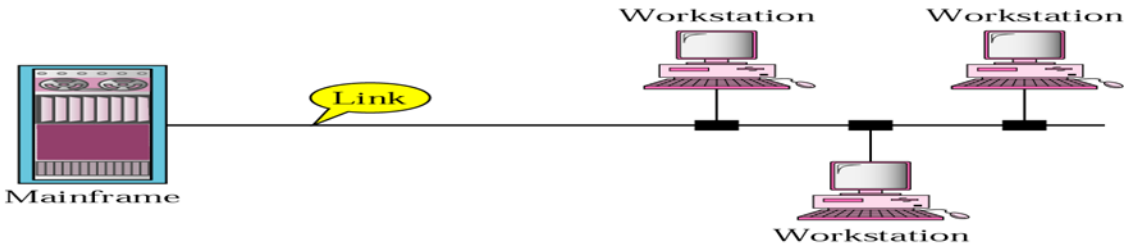
  E,g., Radio wave, Satellite, etc.

Types of connection

- ***Point-to-point:*** *provides a dedicated link between two devices*



- ***Multipoint:*** *more than two devices share (spatially -simultaneously -or temporally by taking turns) a single link*



- ***Direct link:*** *signals propagate directly from transmitter to receiver*
  - *No intermediate devices other than amplifiers or repeaters*
  - *This can apply to both guided and unguided medium*
- ***Indirect link:*** *Systems connected through a switched communication network*

Uses of Computer Networks

a. Business applications

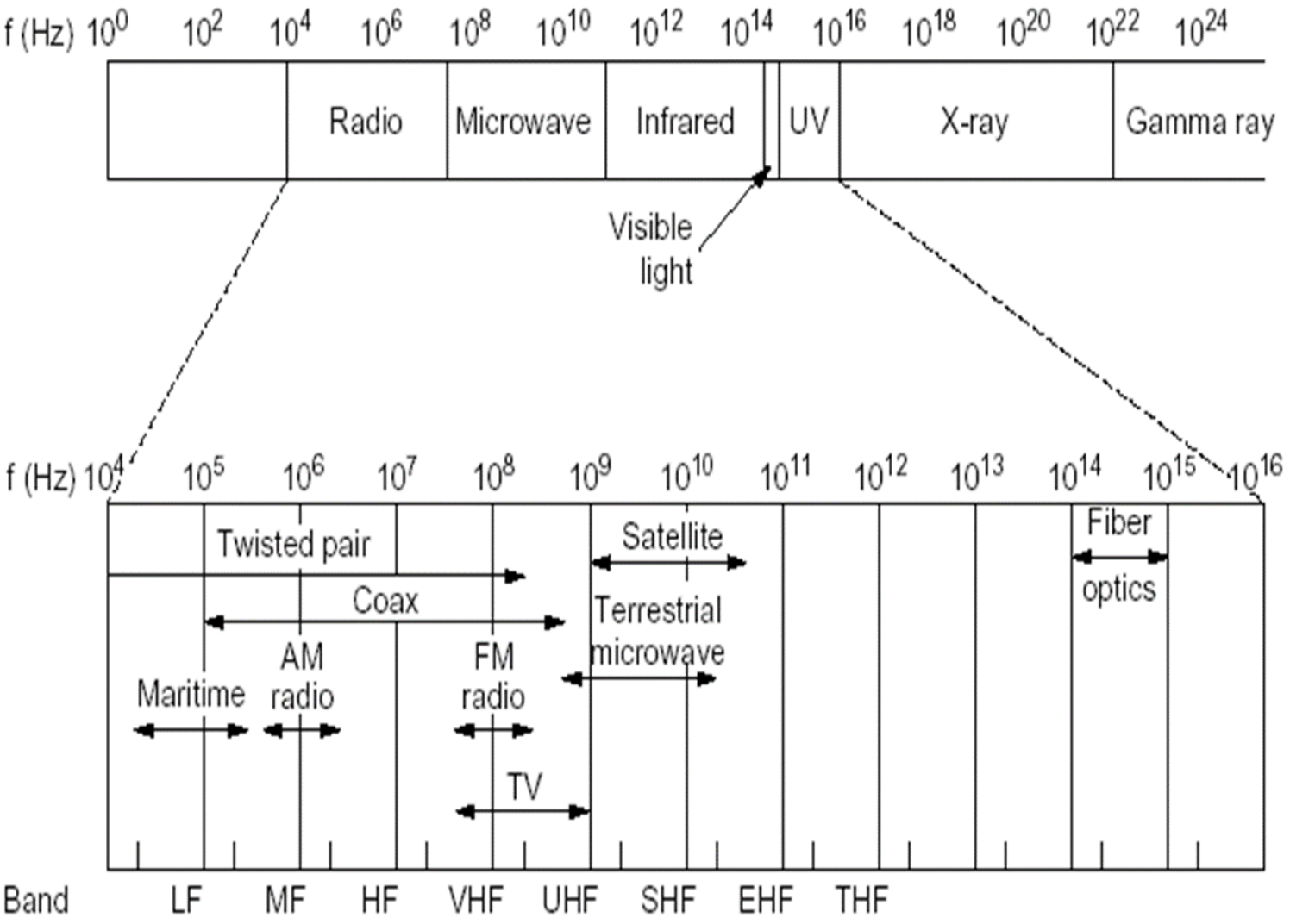
- *for resource sharing including programs, equipment, data (mostly databases on central servers), …*
- *a communication medium – e-mail, writing a report together by making changes on an online document*
- *videoconferencing – to hold meetings by hearing and seeing each other*
- *electronic business*
  - *business to business - placing orders, …*
  - *business with consumers, usually called e-commerce – home shopping*

Everything: government, commercial companies, …

Fundamental of information transmission media $ Encoding

- Data transmission occurs between transmitter and receiver over some transmission medium
- Transmission media may be classified as **guided** or **unguided**
- In both cases communication is in the form of **electromagnetic waves**
- The successful transmission of data depends on
  - the quality of the signal being transmitted
  - the characteristics of the **transmission medium**

Cntd…


f (Hz) 10^0 10^2 10^4 10^6 10^8 10^{10} 10^{12} 10^{14} 10^{16} 10^{18} 10^{20} 10^{22} 10^{24}
Radio
Microwave
Infrared
UV
X-ray
Gamma ray
Visible light
f (Hz) 10^4 10^5 10^6 10^7 10^8 10^9 10^{10} 10^{11} 10^{12} 10^{13} 10^{14} 10^{15} 10^{16}
Twisted pair
Coax
Satellite
Terrestrial microwave
Fiber optics
Maritime
AM radio
FM radio
TV
Band
LF
MF
HF
VHF
UHF
SHF
EHF
THF

Analog and digital data representation

- *In transmitting data from a source to a destination:*
  - *the nature of the data*
  - *the transmission media used to propagate the data*
  - *what processing or adjustments may be required along the way to assure that the received data are understandable*
- *Generally whether we are dealing with **analog** or **digital entities***
- *The term **analog** and **digital** correspond, roughly to continuous and discrete respectively*

Data: *Entities that convey meaning can be*

i. analog: *data take on continuous values on some interval, i.e. continuously varying patterns of intensity.*

E.g. Audio, video, temperature, etc

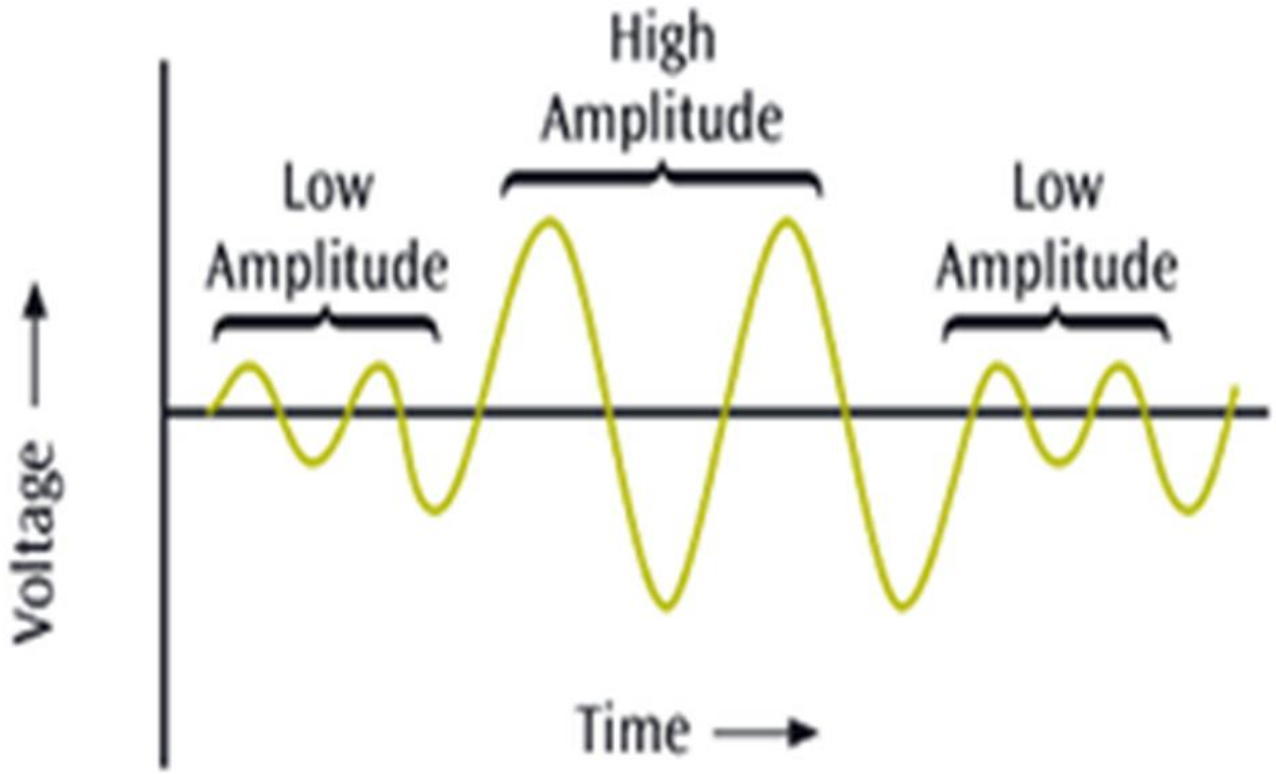
ii. Digital: *data take on discrete values. E.g. text or character strings*

Cntd…

Signals: *Electric or electromagnetic representations of data*

- *Can be also digital or analog*
- ***analog signal*** *is a continuously varying electromagnetic wave that may be propagated over a variety of media .*
- ***Three important characteristics:*** *Amplitude, Frequency, Phase*
  - *Frequency Spectrum of a signal is the collection of all component frequency*
  - *Bandwidth of a signal is the* ***range of component frequencies*** *or the* ***width of the frequency spectrum***
  - *Amplitude: The value of the signal at any point. measured in* ***volts, amperes****, or* ***watts***

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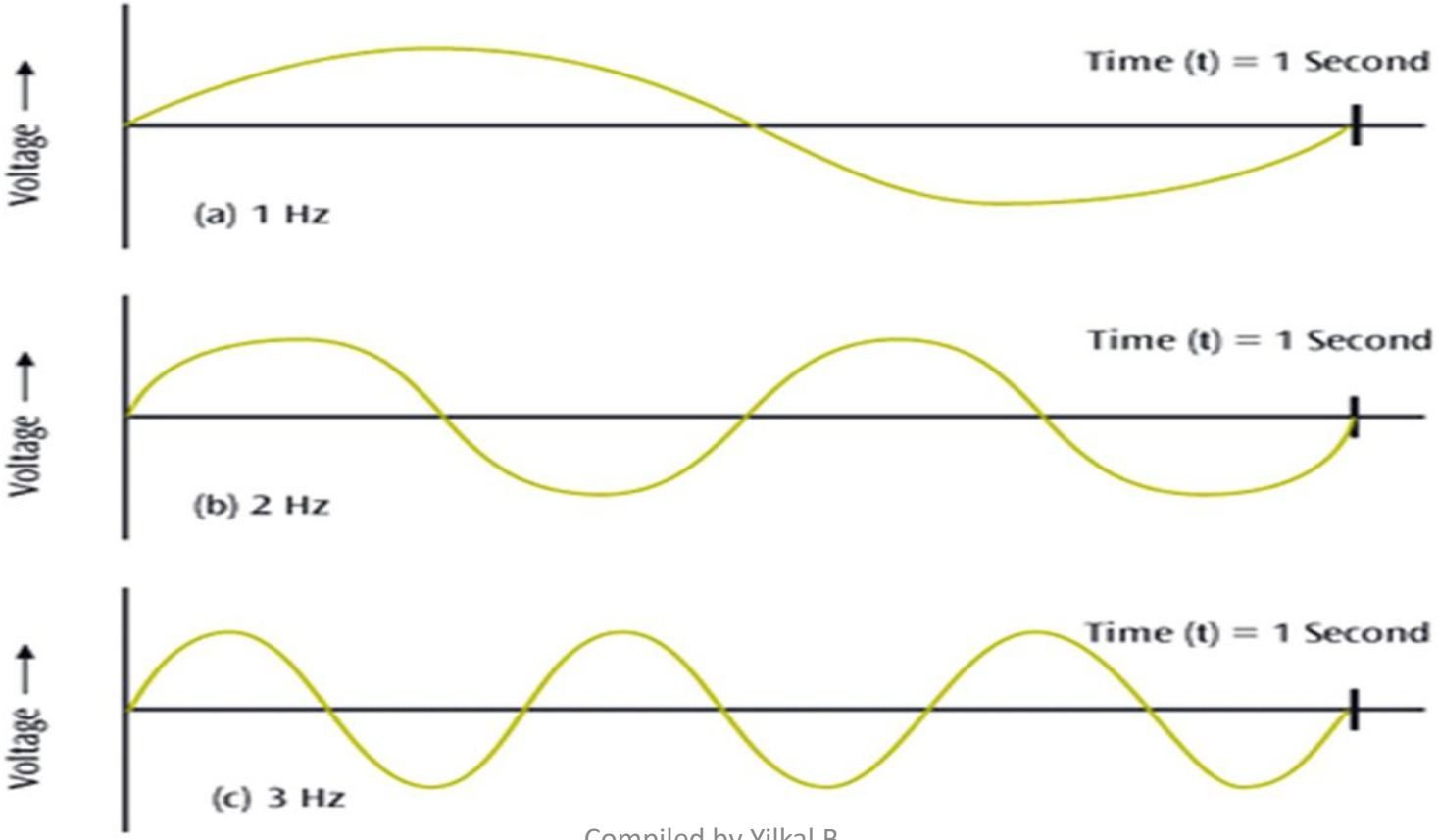


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Frequency : *The number of periods in one second*

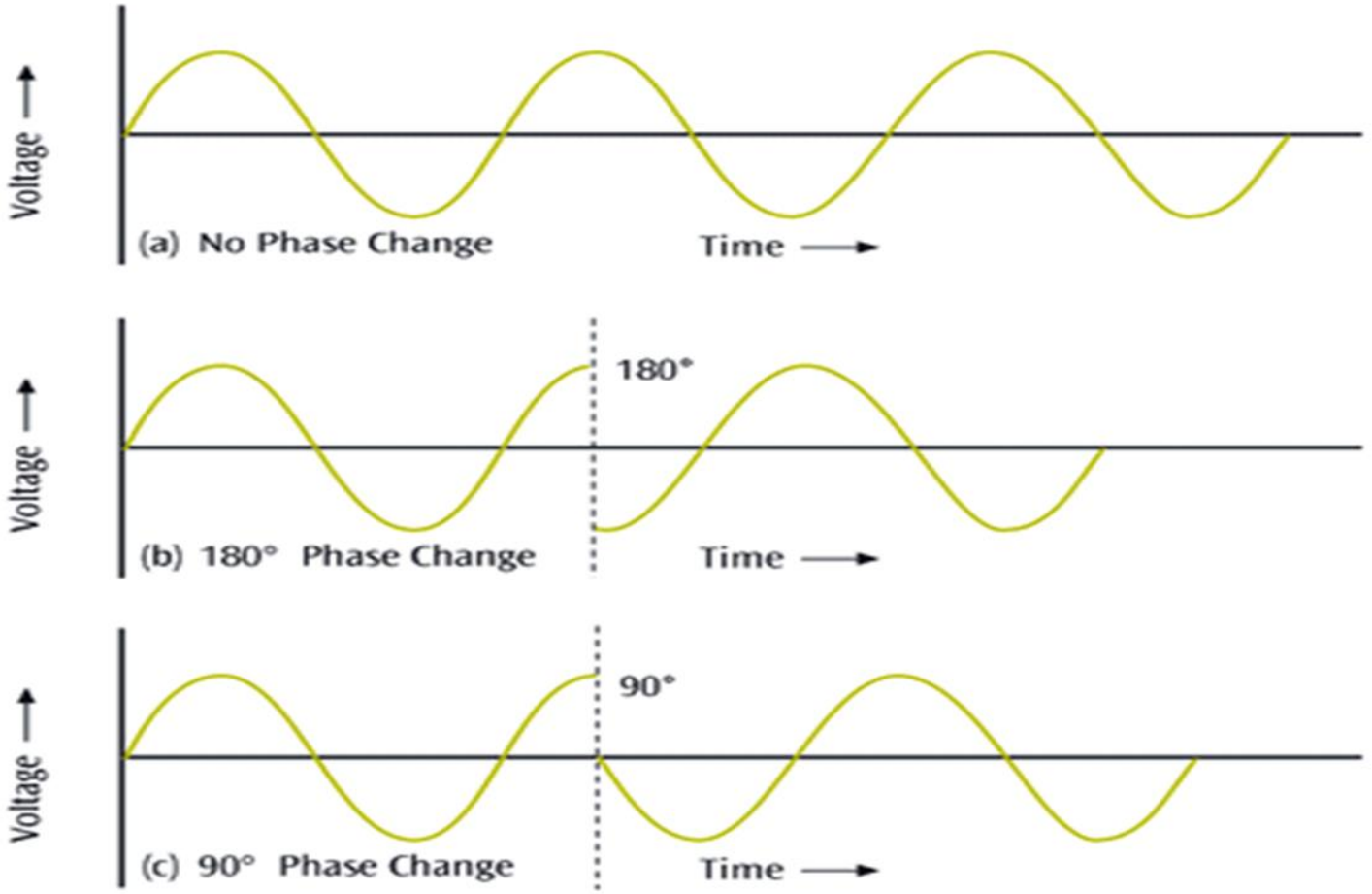
Period : *the amount of time, in seconds, a signal needs to complete one cycle.*

Period and frequency have inverse relationship.



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Phase: *The position of the waveform relative to time zero*



Cntd…

- ***A digital signal*** *is a sequence of voltage pulses that may be transmitted over a wire medium*
- ***Bit Interval:*** *the time required to send a single bit*
- ***Bit rate:*** *the number of* ***bit intervals per second*** *usually expressed in* ***bits per second (bps)***
- *Signaling is the physical propagation of the signal along a suitable medium*

Transmission

- *Communication of data by propagation and processing of signals*
- *The way signals are treated is a function of the transmission system*
- ***Analog transmission*** *is a means of transmitting analog signals without regard to their c ontent the signals may represent analog data (e.g., voice))*
- ***Digital transmission,*** *in contrast, is concerned with* ***the content of the signal.***

Cntd…

- ***Which is the preferred method of transmission?***
  - *Digital transmission is preferable*
- ***The reasons are***
  - ***Digital technology are:***
    - *Continuing to drop in cost and size*
    - *Analog equipment has not shown a similar drop*
  - ***Data integrity***
    - *With the use of **repeaters** the effect of noise and other signal impairments are not cumulative (increasing)*
  - ***Capacity utilization***
    - *High degree of multiplexing*
  - ***Security and privacy***
    - *Encryption techniques can be readily applied to digital data and to analog data that have been digitized*

Transmission Impairments

- *Signal received may differ from signal transmitted*
- ***Analog*** *- degradation of signal quality*
- ***Digital*** *- bit errors*
  - *A binary 1 is transformed into a binary 0 and vice versa*
- ***Caused by***
  - *Attenuation and attenuation distortion*
  - *Delay distortion*
  - *Noise*
- ***Solution***
  - *Amplifiers – analog signal*
  - *Repeaters – digital signal*

Cntd…

Attenuation: *it is a phenomenon which occurs when signal strength* ***falls off*** *with distance*

- *It depends on medium, such as* ***Fiber optic cables*** *carries signal without attenuation up to 2Km.*
- *In communicating entities, received signal strength:*
  - *must be enough to be detected*
  - *must be sufficiently higher than noise to be received without error*

Delay Distortion: *this distortion can only occur in guided media*

- *Propagation* ***velocity varies*** *with frequency*
- *various frequency components of a signal will arrive at the receiver at different times*

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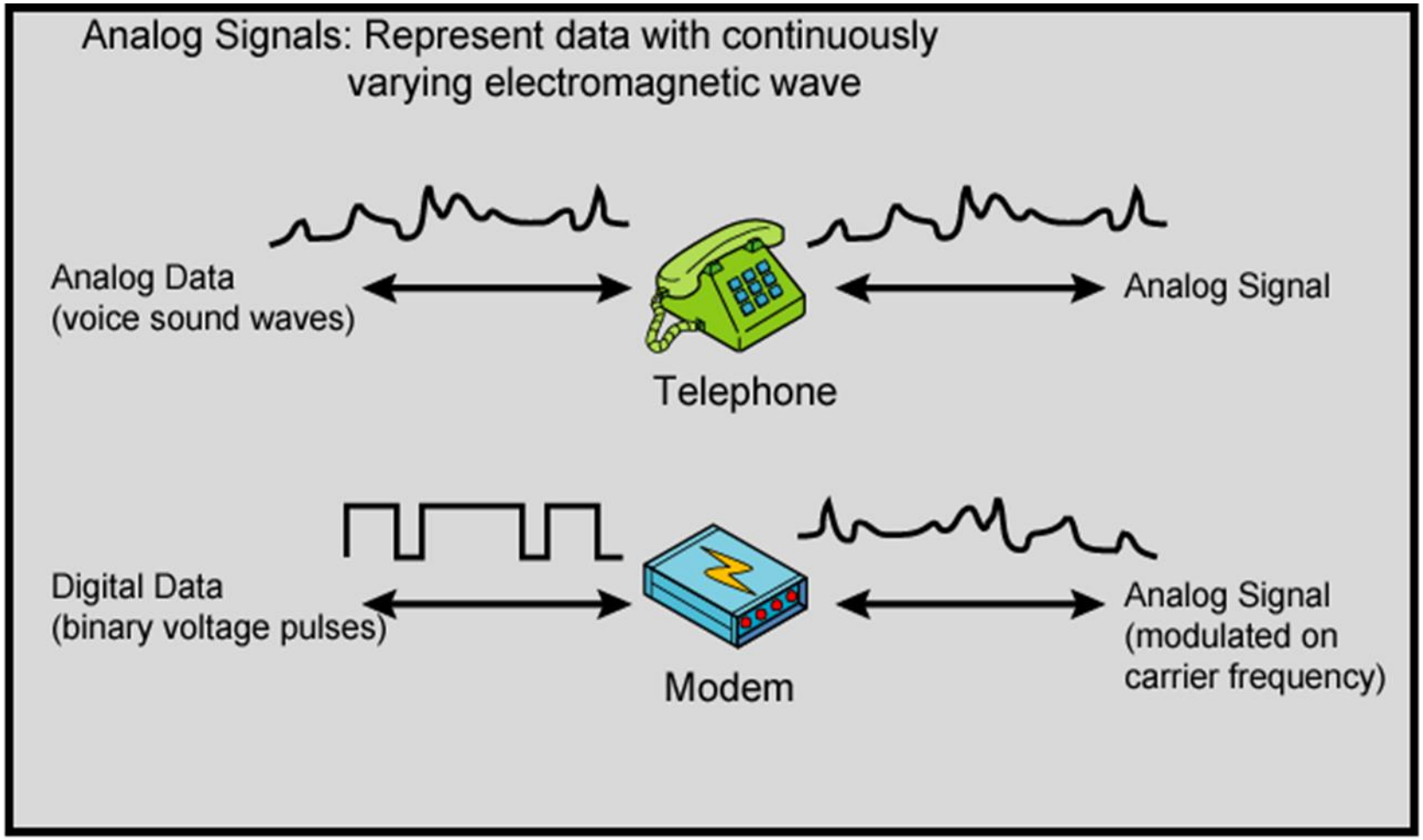
Noise: *additional signals inserted between transmitter and receiver is known as noise*

- *Noises can be caused by*
  - ***Thermal:*** *Due to thermal worry of electrons in a conductor*
  - ***Crosstalk:*** *A signal from one line is picked up by another*
  - ***Impulse Noise*** *: is sharp quick spikes on the signal caused from electromagnetic interference, lightning, sudden power switching, electromechanical switching, etc*
  - ***inter-modulation noise*** *means a noise may occurred during changing from digital to analog or from analog to digital.*

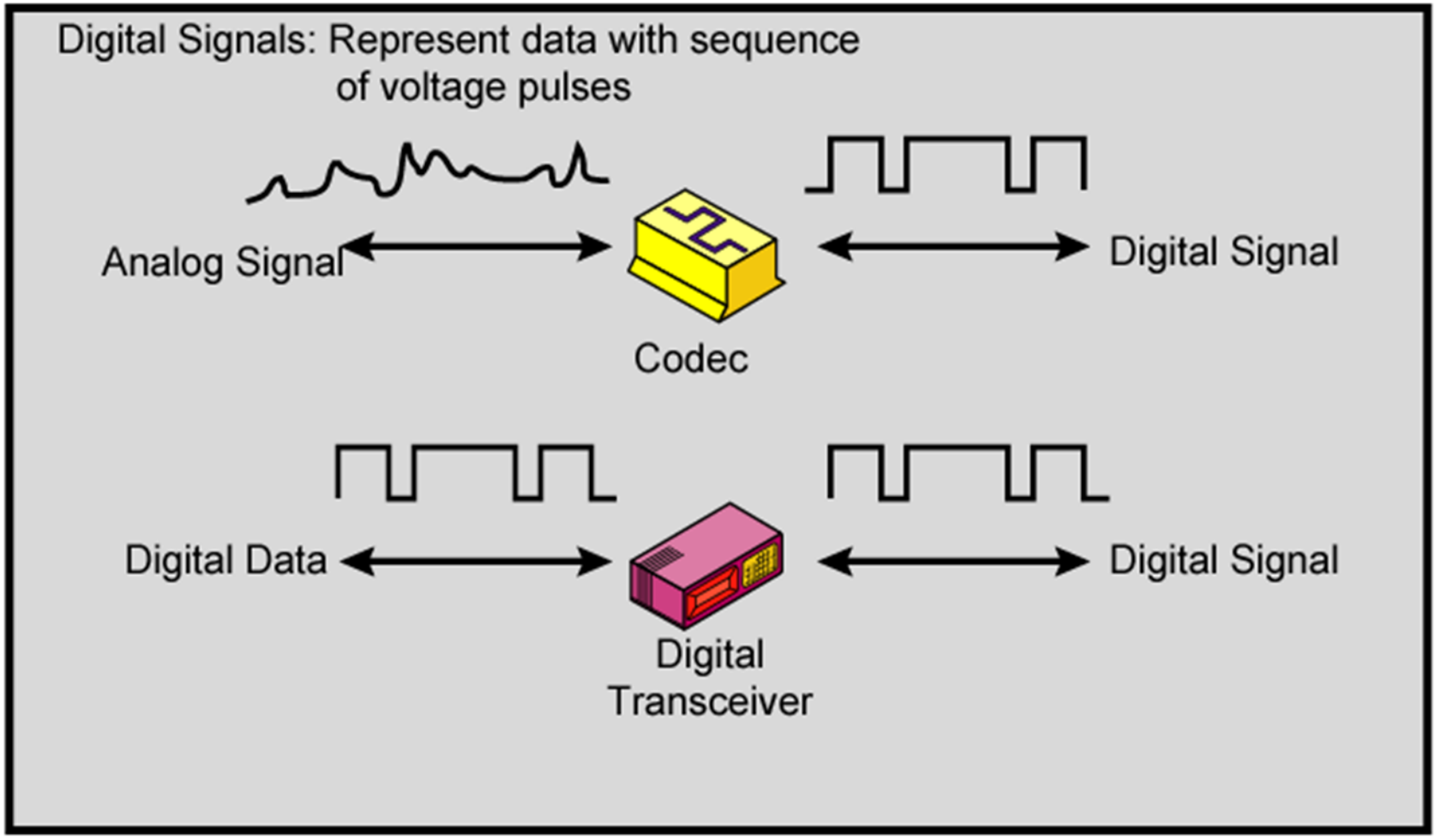
Signal vs data

- ***Analog signals*** *can represent analog and digital data and* ***digital signals*** *can represent digital as well as analog data*
- *Analog data are a function of time and occupy a limited frequency spectrum.*
- *Such data can be represented by an electromagnetic signal occupying the same spectrum*
- ***Digital data*** *can be represented by digital signals*
  - *Different voltage level for each of the two binary digits is used*
- ***Digital data*** *can also be represented by analog signals*
  - *Using modem*
- ***Analog data*** *can be represented by digital signals*
  - *Using codec*

Analog Signals Carrying Analog and Digital Data



Digital Signals Carrying Analog and Digital Data



Encoding and Modulation Techniques

- ***Encoding:*** *the process of preparing data for efficient and accurate transmission*
- ***Modulation:*** *the process of encoding a baseband source signal Sm (t) onto a carrier signal*
- ***Four combinations***
  - *Digital data, digital signal*
  - *Analog data, digital signal*
  - *Digital data, analog signal*
  - *Analog data, analog signal*

Digital Data, Digital Signal:

A digital signal is a sequence of ***discrete****, discontinuous voltage pulses*

- *Each pulse is a signal element.*
- *Binary data are transmitted by encoding each data bit into signal elements*

  E.g. binary 0 is represented by -5V and binary 1 by +5V

- ***At the receiving end the receiver:***
  - *must know the timing of each bit. That is, the receiver must know with some accuracy when a bit begins and ends*
  - *must determine whether the signal level for each bit position is high (1) or low (0)*

Cntd…

- ***Success factors:***
  - *the signal-to-noise ratio*
  - *the data rate*
  - *the bandwidth*
- *The encoding scheme that can be used can also improve performance*
- ***Encoding schemes***

1) ***Non-return to Zero /NRZ***

- *NRZ-L*
- *NRZ-I*

2) ***Bi-phase***

- *Manchester*
- *Differential Manchester*

3) ***Return to Zero (RZ)*** *(**reading Assignment**)*

Cntd….

NRZ: *Uses two different voltage levels (one positive and one negative) as the signal elements for the two binary digits*

- *E.g.* ***Absence of voltage for zero, constant positive voltage for one***
- *More often, negative voltage for one value and positive for the other*
- *Two Variants*
  - *NRZ-L*
  - *NRZ-I*

NRZ-L: *Digital 1s are represented as one voltage (amplitude), while digital 0s are represented as another:*

- *Cheap to implement*
- *Check for voltage of each bit*
- *A long series of 1s or 0s produces a flat, unchanging voltage level (produces synchronization problems)*

Cntd….

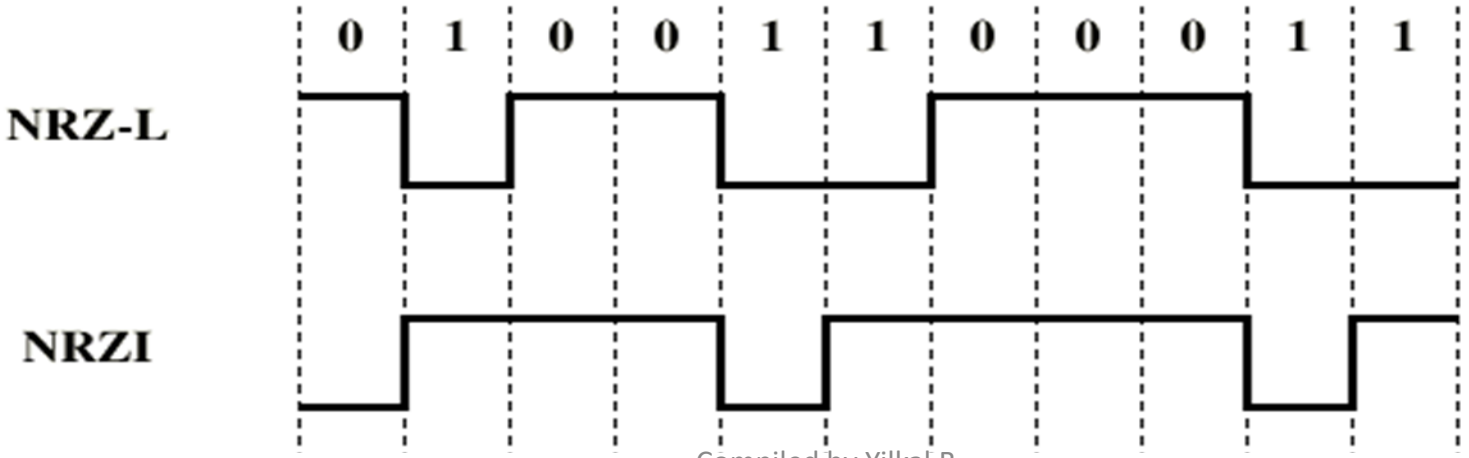
NRZ-I: ***Digital 1s are represented by a voltage change*** *(high-to-low, or low-to-high), while*

0s *are represented as a continuation of the same voltage level:*

- *1 ⇔ existence of a signal transition at the beginning of the bit time (either a low-to-high or a high-to-low transition)*
- *0 ⇔ no signal transition at the beginning of the bit time*

➢ *Even cheaper to implement (only check for changes)*

➢ *A long series of 0s produces a flat, unchanging voltage level*



NRZ Pros and Cons

Pros

- Easy to engineer
- Make good use of bandwidth

Cons

- dc component
- Lack of synchronization capability
- Used for magnetic recording
- Not often used for signal transmission

Biphase

Manchester

- Transition in **middle of each bit period**
- Transition serves as clocking
- Low to high represents one
- High to low represents zero
- Used by IEEE 802.3
  - Used in 802.3 baseband coaxial cable and carrier sense multiple acess / collision detection (CSMA/CD) twisted pair.

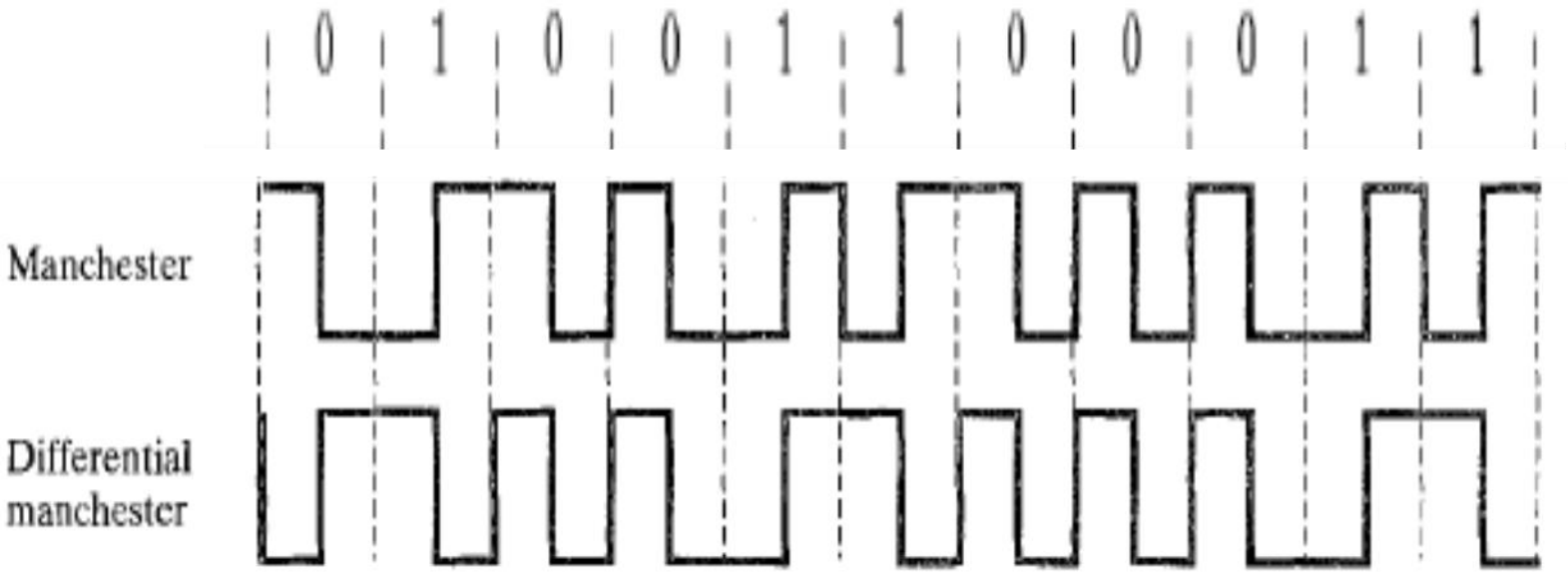
Differential Manchester

- Mid bit transition is clocking only
- Transition at start of a bit period represents **zero**
- No transition at start of a bit period represents **one**

Note: this is a differential encoding scheme

- Used by IEEE 802.5
- Used in 802.5 (token ring) with twisted pair.

Biphase

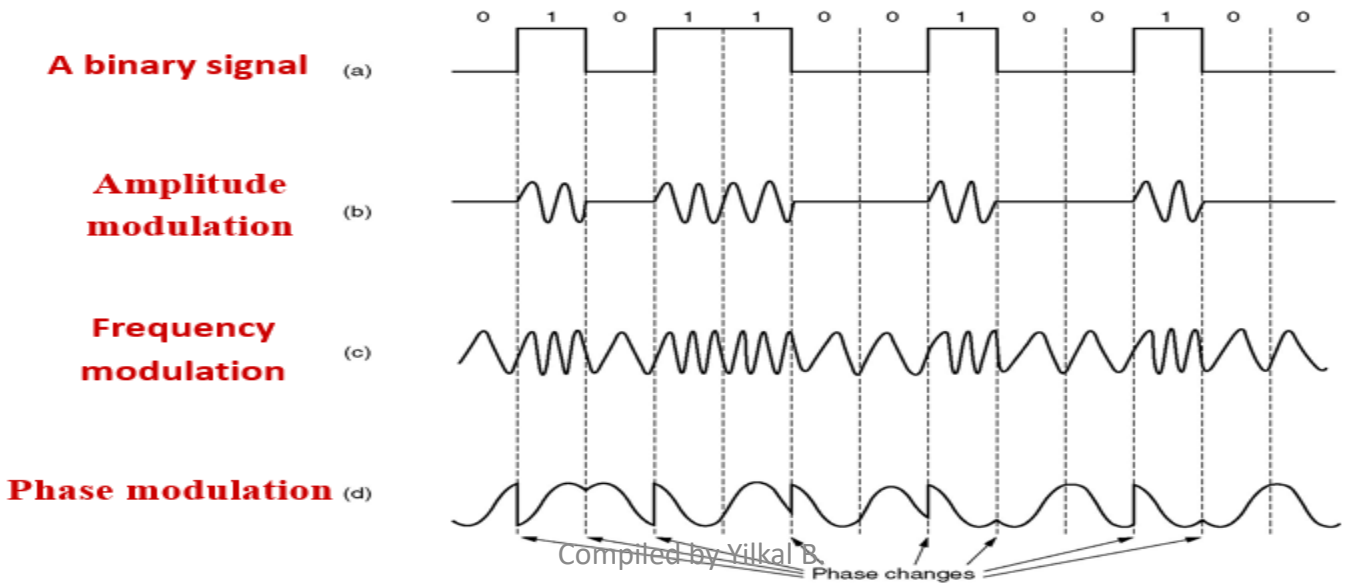


Digital Data, Analog Signals

- The most familiar use of this transformation is for transmitting digital data through the ***public telephone network using*** *modem*
- Basis for analog signaling is a continuous, *constant-frequency signal* known as the ***carrier frequency.***
- Digital data is encoded by modulating one of the *three characteristics* of the carrier: ***amplitude***, ***frequency***, or ***phase*** or some combination of these.
- *there are three basic encoding or modulation techniques*
  - Amplitude-shift keying (ASK)
  - Frequency-shift keying (FSK)
  - Phase-shift keying (PSK)

Cntd…

- ***Amplitude-shift keying (ASK):*** *the two binary values are represented by two different amplitudes of the carrier frequency*
  - *one of the amplitudes is zero the other with value*
- ***Frequency-shift keying (FSK):*** *the two binary values are represented by two different frequencies near the carrier frequency*



Analog Data, Digital Signals

- To send **continuous data** such as **voice** information over a digital transmission medium
- Example: To transfer analog voice signals off a local loop to digital end office within the phone system, one uses a **codec**
- The most common technique for using digital signals to encode analog data is **Pulse Code Modulation(PCM)**

Pulse Code Modulation(PCM)

Steps

- sample the continuous data - to measure the value of the continuous data at equal interval
  - Each sample is called **Pulse**
- the resulting pulse is **quantized** - assigned a value
- each value is transformed to assign-and-magnitude binary number

Analog Data, Analog Signal

- *There are **two principal reasons***
  - *A higher frequency may be needed for effective transmission*
  - *Modulation permits frequency-division multiplexing*
- *The principal techniques for modulation using analog data are:*
  - *amplitude modulation (AM)*
  - *frequency modulation (FM)*
  - *phase modulation (PM)*

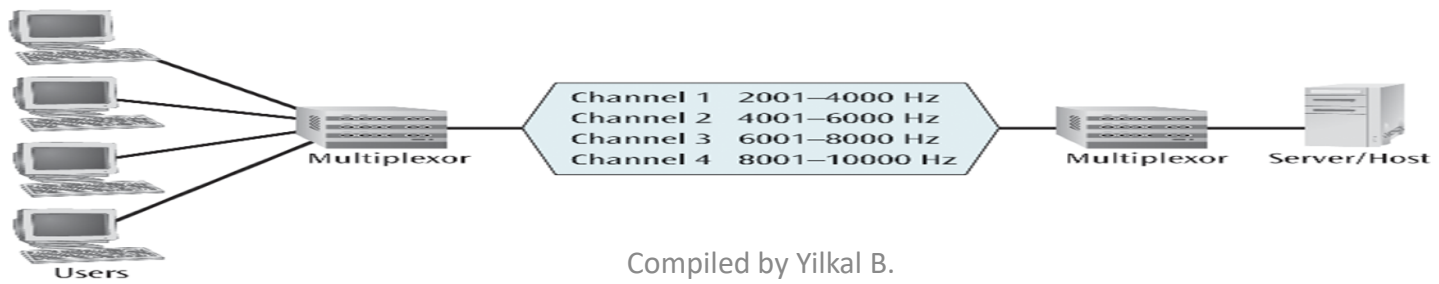
Multiplexing

- *term used to refer to a process where **multiple** analog message signals or digital data streams are combined **into one signal** over a shared medium*
- *to share an expensive resource*
- *multiplexing divides the capacity of the low-level communication channel into several higher-level logical channels, one for each message signal or data stream to be transferred*
- *A reverse process, known as **demultiplexing**, can extract the original channels on the receiver side*
- *The two most basic forms of multiplexing are :*
  - *Time-division multiplexing (TDM) –for digital signal*
  - *Frequency-division multiplexing (FDM) –for analog signal*

Cntd…

FDM

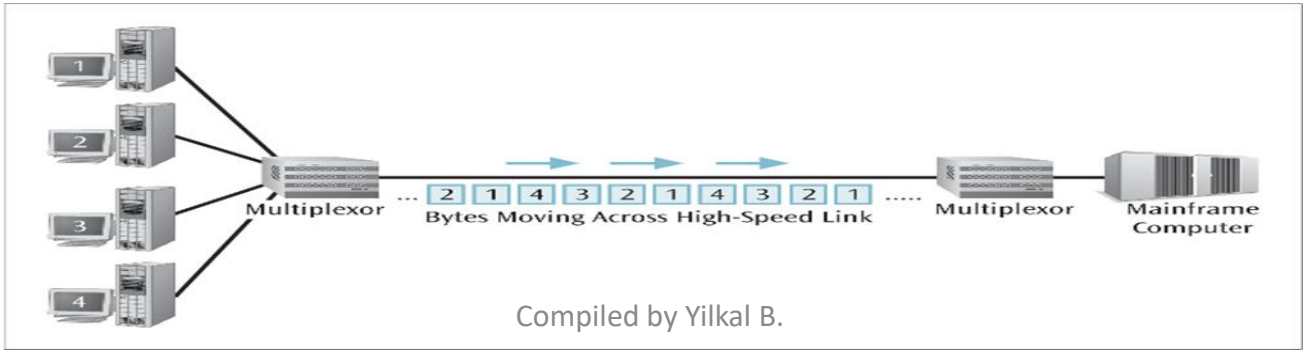
- The frequency spectrum is divided into frequency bands, with each user having exclusive possession of some band
- possible when the useful bandwidth of the transmission medium exceeds the required bandwidth of signals to be transmitted
- the frequency-division multiplexing (FDM) used for analog signals.
- A number of signals can be carried simultaneously if each signal is modulated onto a different carrier frequency
- carrier frequencies are sufficiently separated that the bandwidths of the signals do not overlap



Cntd…

TDM

- The users take turns, each one periodically getting the entire bandwidth for a little burst of time.
- time-division multiplexing (TDM) is used for digital signals
- With TDM, there is no inter-modulation noise, whereas we have seen that this is a concern for FDM.
- possible when the achievable data rate (sometimes, called bandwidth) of the medium exceeds the data rate of digital signals to be transmitted
- Multiple digital signals (or analog signals carrying digital data) can be carried on a single transmission path by interleaving portions of each signal in time.



End of chapter one

Any question???